



# Biomolecules - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

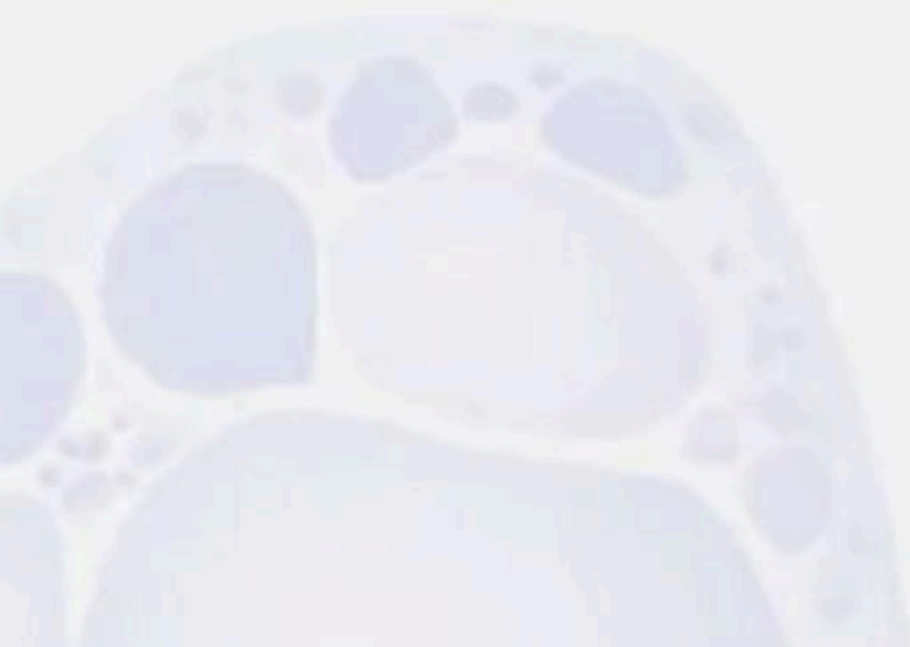


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# Biomolecules

100 M (G)



**BIOLOGYLIVE**

## Next class :-

22<sup>nd</sup> Jan → "Plant Growth and Development" (6PM)

[23<sup>rd</sup> Jan → Photosynthesis (10AM) ✓  
23<sup>rd</sup> Jan → Breathing (6PM) ✓]

[24<sup>th</sup> Jan → Respiration (10AM)  
24<sup>th</sup> Jan → Body Fluid (6PM)]

[Note: "2 days" No  
Session but i'll  
Give MCQs and  
Sheets so do solve.]



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1. Imp  
Match the following :

202s

a. Vinblastin

b. Ricin

c. Codeine

d. Gum

i. Alkaloid

ii. Polymer

iii. Drug

iv. Poisonous

(1) a-iii, b-iv, c- ii, d- i

(2) a-i, b-ii, c- iv, d- iii

✓✓ (3) a-iii, b-iv, c- i, d- ii

(4) a-iv, b-iii, c- i, d- iv



2. Which of the following are polymeric :

- (1) Proteins ✓
- (2) Polysaccharides ✓
- (3) Nucleic acids ✓
- (4) All of these ✓





3. Given below are two statements :

**Statement I :** Molecules in the insoluble fraction with exception of lipids are polymeric substance.

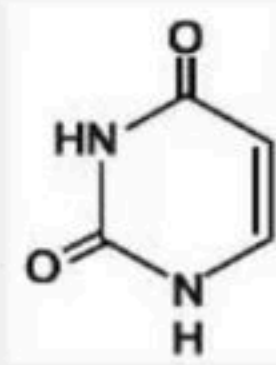
**Statement-II :** Some secondary metabolites have ecological importance.

**Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



4. Which of the following is correct according to given diagrammatic representation :



Uracil = only RNA

- (1) It is a nucleoside present in RNA
- (2) It is a nitrogenous base present in DNA
- ✓ (3) It is a nitrogenous base present in RNA
- (4) It is a nitrogenous base present in DNA and RNA





### 5. Inulin consist of :

(1) Only glucose

✓ (2) Only fructose

(3) Only maltose

(4) Protein



6. Which of the following is true :

polymer of sugar

(1) Polysaccharides are starting part of sugar chain ~~X~~

✓ (2) Cellulose is homopolymer

(3) The starch- $I_2$ , is ~~red~~ in colour

blue

(4) All of these



7. A  $\beta$ -plated sheet organisation in a polypeptide chain is an example of :

(1) 1° structure

✓✓ (2) 2° structure

(3) 3° structure

(4) 4° structure





8. Which of the following is not true :

(1) In B DNA, the number of base pair is 10 ✓

✓ (2) The pitch of DNA is 3.4 Å ~~x~~ 34 Å

(3) There are two hydrogen bond between A and T  $A=T$

(4) There are three hydrogen bond between G and C  $G=C$



9. Given below are two statements :

**Statement I :** The living state is a non equilibrium steady state to be able perform work.

**Statement-II :** Living state and metabolism are synonymous → Similar

**Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct





10. Choose the type of enzyme involved in the following reaction.

2025



(1) Dehydrogenase

✓ (2) Transferase

(3) Hydrolase

(4) Lygase





11. Find correct matching is/are :

(a) Watson – Crick model = Secondary structure ✓

(b) Pocket of enzyme = 'Active site' ✓

(c) Trypsin = ~~hormone~~ Enzyme (protein) ✓

(1) a and c only

✓ (2) a and b only

(3) a only

(4) a, b and c



v.v.v.m?

12. Zinc is a cofactor for the ..... enzyme :

- ✓✓ (1) Proteolytic
- (2) Transferase
- (3) Lyases
- (4) Ligases



**13. Chitinous exoskelton of arthrodpoes is formed by the polymerisation of :**

- ✓ (1) N – acetyl glucosamine
- (2) Lipiglycans
- (3) D – Glucosamine
- (4) Keraitin sulphate





**14. Given below are two statements :**

**Statement I :** Metabolism provides a mechanism for the production of energy ✓

**Statement II :** Living organism exist in an equilibrium state. ✗

**Choose the correct answer from the options given below.**

- (1) Both Statement I and Statement II are incorrect
- ✓ (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct



**15. Given below are two statements :**

**Statement I :** Each of metabolic reactions results in transformations of biomolecules ✓

**Statement II :** Protein with catalytic power are named enzymes ✓

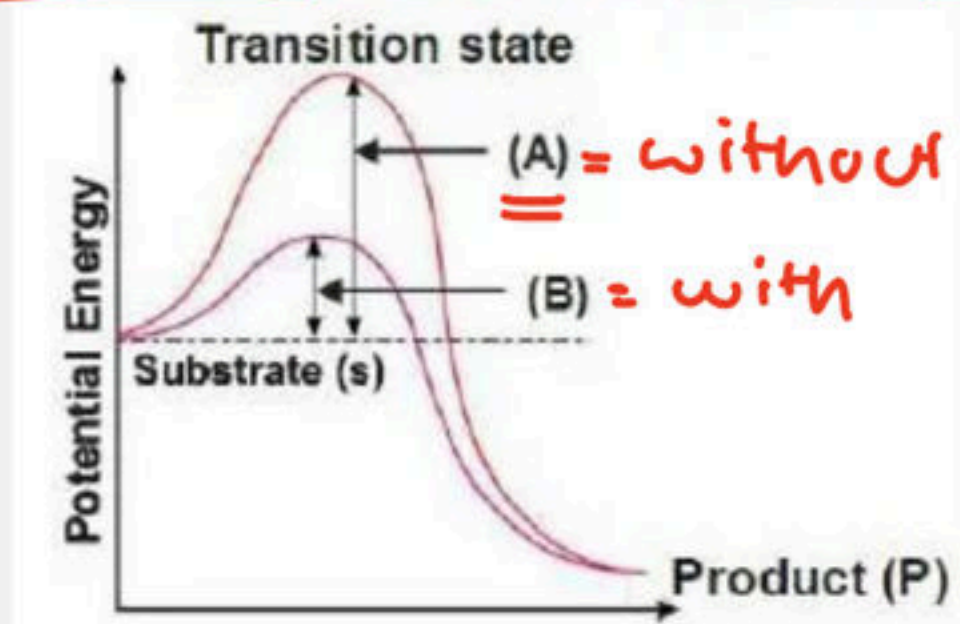
**Choose the correct answer from the options given below.**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ✓(4) Both, Statement I and Statement II are correct





16. In the given graphs which graphs represents, activation energy without enzyme and with enzyme. Choose the correct option :



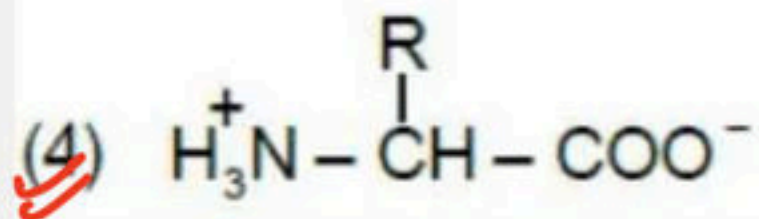
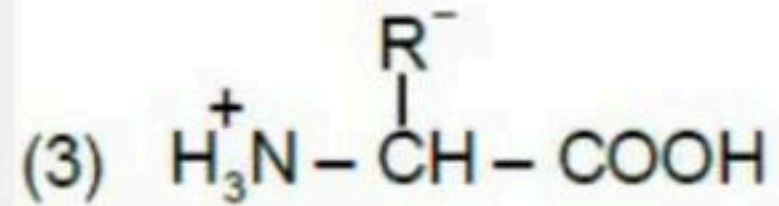
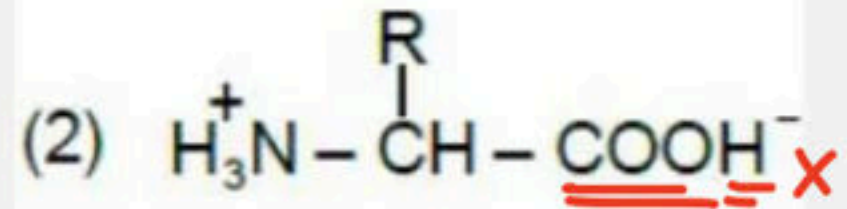
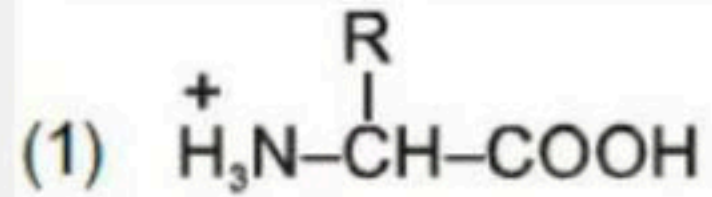
PYQs

- (1) A = Activation energy with enzyme
- ✓ (2) B = Activation energy with enzyme
- (3) B = Activation energy without enzyme
- (4) 1 and 3 both





17. Which of the following is zwitterion : 2023



18. Prosthetic group is part of an enzyme it is : 2025

- (1) Loosely attached organic compound
- (2) Tightly attached inorganic compound
- ✓ (3) Tightly attached organic part
- (4) Loosely attached inorganic compound



### 19. What are pockets in an enzyme :

- (1) Crevices made due to chain criss - crossing in tertiary structure of protein ✓
- (2) An active sites of an enzyme ✓
- (3) Site where substrates fit ✓
- (4) All of these ✓





20. Match the following column I and column II : 2025

Column I

a. Acidic amino acid

b. Aromatic amino acid

c. Neutral amino acid

d. Basic amino acid

Column II

i. Lysine

ii. Valine

iii. Glutamic acid

iv. Tyrosine

✓ (1) a-iii, b-iv, c-ii, d-i

(3) a-i, b-iv, c-iii, d-ii

(2) a-ii, b-iii, c-iv, d-i

(4) a-iv, b-iii, c-i, d-ii



21. Which one of the following pairs of nitrogenous bases of nucleic acids, is wrongly matched with the category mentioned against it :

- (1) Guanine, adenine–purines
- ✓ (2) Adenine, thymine–purines
- (3) Thymine, uracil–pyrimidines
- (4) Uracil, cytosine–pyrimidines

AG = Puri

CUT = Pyrid





## 22. Haemoglobin consists of :

- ☒ (1) 2 $\alpha$  and 2 $\beta$  subunits
- (2) 3 $\alpha$  and 3 $\beta$  subunits
- (3) 4 $\alpha$  and 4 $\beta$  subunits
- (4) 1 $\alpha$  and 1 $\beta$  subunits





23. In a certain process a molecule of glucose converts into lactic acid in skeletal muscle. Can you tell which type of process is this :

- (1) Anabolic
- ✓ (2) Catabolic
- (3) Amphibolic
- (4) All of these



24. Almost all enzymes are proteins and are composed of one or several polypeptide chains. However there are a number of cases in which non-protein constituents called as :

(1) Apoenzyme

✓ (2) Co-factor

(3) Allosteric

(4) Holoenzyme



25. A typical <sup>→ TGA</sup>fat molecule is made up of :

- (1) One glycerol and one fatty acid molecule
- (2) Three glycerol and three fatty acid molecules
- (3) Three glycerol molecules and one fatty acid molecule
- (4) One glycerol and three fatty acid molecules.





## 26. Macro molecule chitin is :

- (1) Sulphur containing polysaccharide
- (2) Simple polysaccharide
- ✓ (3) Nitrogen containing polysaccharide
- (4) Phosphorus containing polysaccharide



27. Which one of the following is a non - reducing carbohydrate :

(1) Maltose

202 s

✓ (2) Sucrose

(3) Lactose

(4) Ribose 5 - phosphate





28. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

✓ **Assertion (A)** : Molecular weight of lipid do not exceed 800 Da but it come under acid insoluble fraction. ↑ ✓

✓ **Reason (R)** : When we grind tissue cell membrane and other membrane are broken into pieces and form vesicles which are not water soluble.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- ✓ (4) Both (A) and (R) are true and (R) is the correction explanation of (A)





29. Transition state structure of the substrate formed during an enzymatic reaction is :

- ✓ (1) Transient and unstable
- (2) Permanent and stable
- (3) Transient but stable
- (4) Permanent but unstable



30. The reaction is catalysed by



2025

(1) Transferases

(2) Hydrolases

✓ (3) Lyases ✓

(4) Isomerases



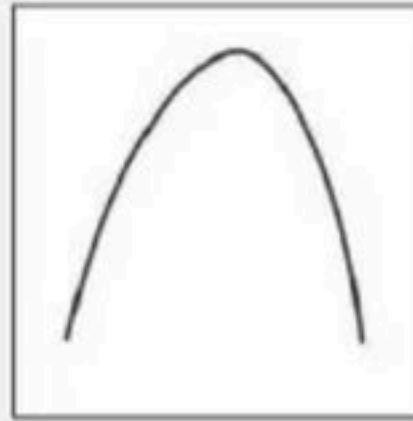
31. DNA and RNA are similar in having :

- (1) Similar pyrimidines
- (2) Similar sugar
- ✓ (3) Both are polymers of nucleotides
- (4) Both are double stranded





32. If Y axis is enzymetic activity shown in the above graph, then X axis can be :



- (1) pH
- (2) Temperature
- (3) Concentration of substrate
- ✓ (4) Both (1) and (2)



33. Given below are two statements :

✓ **Statement I** : Biomolecules which have molecular weight less than one thousand dalton are usually referred to as micromolecules.

✗ **Statement-II** : Proteins, nucleic acid, polysacharide and lipids have molecular weight in range of ten thousand daltons and above.

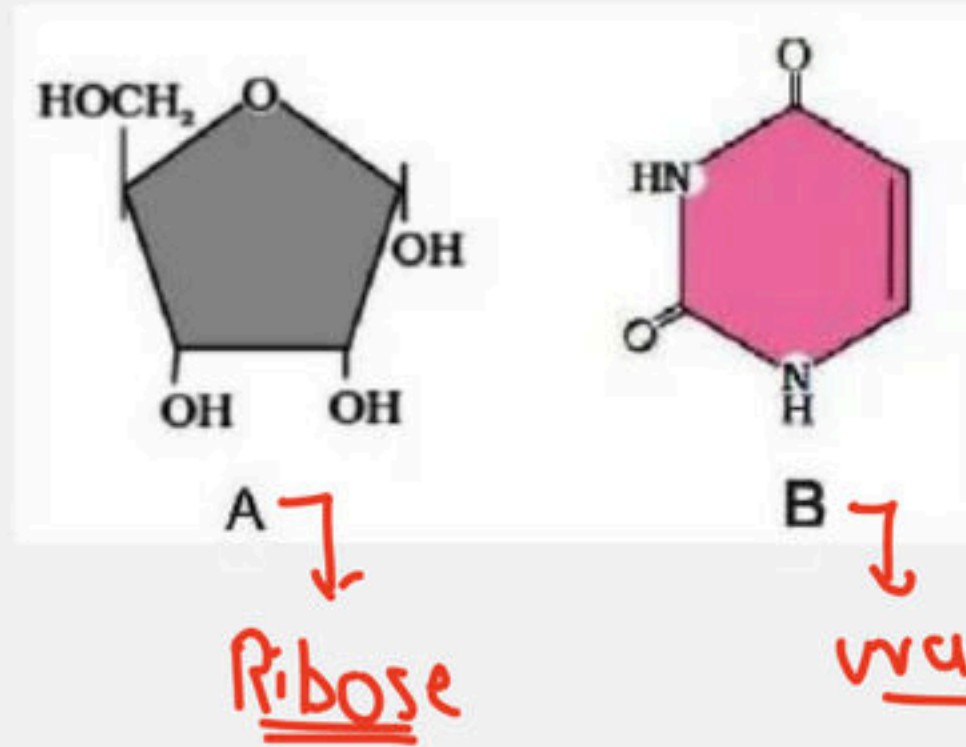
**Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- ✓ (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct





34. The given diagram A and B respectively



- ✓ (1) Ribose and uracil
- (2) Fructose and thymine
- (3) Glucose and pyrimidine
- (4) Ribose and purine





35. Which of the following incorrect option :

(1) Adenine, Guanine, Cytosine, Thymine → Nitrogen base

(2) Nitrogen base + Sugar → Nucleoside

(3) Nitrogen base + Sugar + Phosphate → Nucleotide

✓ (4) Nitrogen base + Phosphate → Nucleoside  
+ Sugar



36 "Non-nitrogenous" biomolecules among the following are:

✓ I. Polysaccharides

II. Nucleic acids →

III. Proteins →

✓ IV. Lipids

(1) I only

(2) I and II only

✓ (3) I and IV only

(4) I, III, and IV



37. Which one of the following cannot be attributed to glycogen?

(1) Homopolysaccharide = Glycogen

☒ (2) Heteropolysaccharide

(3) Branched chain molecule

(4) Stored in liver and muscle





38. The helical structure of proteins is stabilized by:

(1) Dipeptide bonds

✓ (2) Hydrogen bonds

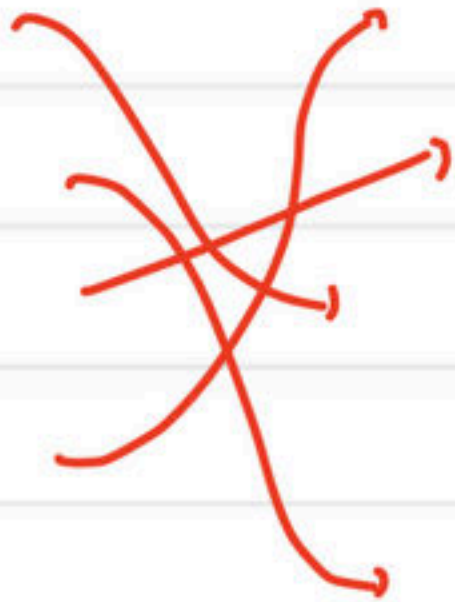
(3) Ether bonds

(4) Peptide bonds



39. Match the columns and find out the correct combination:

| Column A    | Column B                       |
|-------------|--------------------------------|
| A. Morphine | 1. Toxin                       |
| B. GLUT-4   | 2. Drug                        |
| C. Curcumin | 3. Alkaloid                    |
| D. Ricin    | 4. Sensory receptor            |
|             | 5. Glucose transport (Protein) |



Options:

(1) A-3, B-5, C-2, D-4

(3) A-2, B-3, C-1, D-4

✓ (2) A-3, B-5, C-2, D-1

(4) A-3, B-1, C-2, D-4



✓1. ✓.1mp

40. Oils remain as liquid in winters because:

- (1) They have a lower melting point
- (2) They contain unsaturated hydrocarbons
- (3) They have single carbon-carbon bonds
- (4) Both (1) and (2)





## 41. Which statement regarding enzyme inhibition is correct?

- (1) Competitive inhibition occurs when a substrate competes with an enzyme for binding to the inhibitor protein.
- ✓ (2) Competitive inhibition occurs when the substrate and the inhibitor compete for the active site of the enzyme.
- (3) Non-competitive inhibition of an enzyme can be overcome by adding a large amount of substrate.
- (4) Non-competitive inhibitors often bind to the enzyme irreversibly.



42. Match the columns and find out the correct combination:

2025

| Protein     | Function                          |
|-------------|-----------------------------------|
| A. Insulin  | 1. Sensory reception              |
| B. Collagen | 2. Hormone                        |
| C. Receptor | 3. Biocatalyst                    |
| D. Trypsin  | 4. Enzyme                         |
|             | 5. Intercellular Ground substance |

Options:

(1) A-2, B-5, C-3, D-4

(3) A-3, B-5, C-1, D-4

(2) A-2, B-4, C-1, D-5

☒ (4) A-2, B-5, C-1, D-4





43. Match the columns and find out the correct combination: 2025

| Component        | % of total cellular mass       |
|------------------|--------------------------------|
| A. Lipids        | 1.1                            |
| B. Nucleic acids | <del>2</del> 2                 |
| C. Protein       | 3. 5-7                         |
| D. Ions          | 4. <u>70-90</u> = <u>water</u> |
|                  | 5. 10-15                       |

Options:

(1) A-2, B-5, C-3, D-1

☒ (2) A-2, B-3, C-5, D-1

(3) A-3, B-2, C-5, D-1

(4) A-5, B-2, C-1, D-4





44. Which one of the following statements is correct with reference to enzymes? 2025

- (1) Apoenzyme = Holoenzyme + Coenzyme
- ✓ (2) Holoenzyme = Apoenzyme<sup>Protein</sup> + Coenzyme<sup>No protein</sup>
- (3) Coenzyme = Apoenzyme + Holoenzyme
- (4) Holoenzyme = Coenzyme + Cofactor



45. Which of the following are not polymeric?

(1) Nucleic acids

(2) Proteins

(3) Polysaccharides

✓ (4) Lipids



46. A metal ion required for normal functioning of an enzyme is:

- (1) Holoenzyme
- (2) Coenzyme
- ☒ (3) Cofactor
- (4) Prosthetic Group





**47. NAD stands for:**

- (1) Nicotinamide adenosine diphosphate
- (2) Nicotine adenosine phosphate
- ✓ (3) Nicotinamide adenine dinucleotide
- (4) None of the above



48. If 'P' (product) is at a lower level than 'S' (substrate), the reaction is an:

- (1) Exothermic reaction ✓
- (2) Endothermic reaction
- (3) Spontaneous reaction ✓
- (4) ✓ Both (1) and (3)



## 49. Enzymes enhance the rate of reaction by:

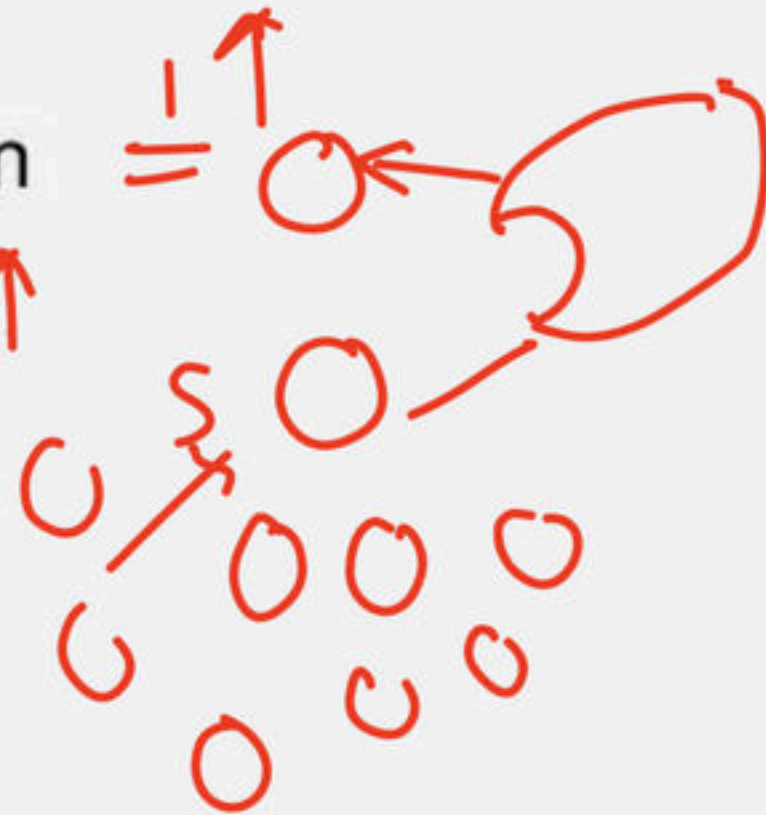
- (1) Combining with the product
- (2) Forming reactant-product complex
- (3) Charging the equilibrium of the reaction
- (4) Lowering activation energy





50. The effect of a competitive inhibitor can be reversed by: NEET 2025

- (1) Increasing the inhibitor concentration
- (2) Decreasing the substrate concentration
- ✓ (3) Increasing the substrate concentration ↑
- (4) The effect is irreversible



51. Which of the following incorrect option:

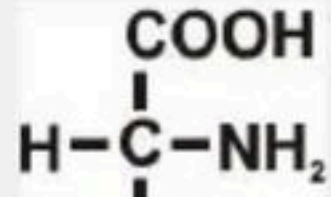
(1) Carotenoids and anthocyanin are secondary metabolites.

(2) Carotenoids and anthocyanin → Pigment

✓ (3) Rubber, Gum, Cellulose → ~~Drugs~~

(4) Rubber, Gum, Cellulose → Polymeric substance





52. R-CH<sub>3</sub> This amino acids is :

(1) Glycine

(2) Serine

✓ (3) Alanine

(4) Arginine

R-H = Glycine





53. Which of the following is not the primary metabolites :

- (1) Glucose
- (2) Amino acid
- (3) Fatty acid
- ✓ (4) Ricin



54. Match the column :

## Column A

i- Collagen

ii- GLUT - 4

iii- Insulin

iv- Receptor

## Column B

a- Hormone

b- Inter cellular ground substances

c- Sensory receptor

d- Enables glucose transport into cells

✓✓ (1) i-b, ii-d, iii-a, iv-c

(3) i-d, ii-b, iii-a, iv-c

(2) i-b, ii-d, iii-c, iv-a

(4) i-c, ii-a, iii-d, iv-b





55. Which of the following statements is/are correct:

- (i) Tertiary structure is neccessary for biological activity of protein ✓
- (ii) Nucleic acids exhibit a wide variety of secondary structures ✓
- (iii) Haemoglobin is an example of quaternary structure
- (iv) In primary structure of protein the first amino acid is called as N-terminal amino acid and the last amino acid is called C-terminal amino acid

Stul 2025 J

(1) ii only

(2) i and iv only

(3) i only

✓✓ (4) i, ii, iii and iv





56. Correct for DNA double helical structure of B-DNA:

(a) Helical stair of ascent is – ~~30°~~ 36°

(b) Turn of helical strand involve – 10 step ✓

(c) Distance between two successive base pair is – 3.4 Å ✓

(d) Pitch of helix is – ~~38 Å~~ 34 Å

(1) a, b, c, d

✓ (2) Only b, c

(3) Only a, b, d

(4) Only c, d



57. Which one of the following correct statement :

- (a) Prosthetic group are organic compound ✓
- (b) Proteineous portion of enzyme is called apoenzyme ✓
- (c)  $\text{Zn}^{+2}$  is a co-factor for the proteolytic enzyme carboxypeptidase ✓
- (d) NAD and FAD are Co-factor ✓

(1) Only a, b

(2) Only a, d

(3) Only c, d

(4) a, b, c and d ✓





58. Enzyme which catalyses the linking together of two compounds is :

(1) Hydrogenase

(2) Transferase

✓✓ (3) Ligases

(4) Lyases.





59. If a living tissue is grinded in trichloroacetic acid in pestle and mortar and then filtered through cheesecloth, the acid soluble portion will come in

- ✓ (1) Filtrate ↗
- (2) Retentate
- (3) No acid soluble portion will be there
- (4) Both 2 and 3



60. Which of the following are aromatic amino acids

(a) Tryptophan

(b) Tyrosine

(c) Phenylalanine

(d) Histidine ~~x~~

(1) b and c only

☒ (2) a, b, c only

(3) a and b only

(4) all of these



61. Palmitic and arachidonic acid have how many carbons including carboxyl carbon

↑ 16C

↓  
20C

(1) 16,18

✓ (2) 16,20

(3) 15,19 [excludina]

(4) 15,20





**62. In lipids glycerol and fatty acids are joined by**

- ✓(1) Esterification
- (2) Glycosidic linkage ✗
- (3) Aldehyde
- (4) Peptide bond



63. Phosphate group is attached to nucleoside by

- (1) Phosphoester bond
- (2) Glycosidic bond
- (3) Ether bond
- (4) Both 1 and 2



**64. Proteins are generally :**

(1) Homo polymer

✓ (2) Heteropolymer ✓

(3) Monomer

(4) None of these





**65. Source of essential amino acids are**

(1) Synthesised in body

☒ (2) Through diet

(3) Both 1 and 2

(4) None of these



66. "Ramachandran plot" is used to confirm the structure of :

(1) RNA

✓ (2) Proteins

(3) Triachylglycerides

(4) DNA



**67. A non-proteinaceous enzyme is**

(1) Lysozyme

✓ (2) Ribozyme

(3) Ligase

(4) Deoxyribonuclease.





68. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A) :** Nonprotein constituent called cofactor are bound to enzyme to make the enzyme catalytically active. ✓

**Reason (R) :** Only co-enzyme serve as cofactor in enzyme catalyzed reaction. ✗

In the light of the above statements, choose the correct answer from the options given below

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

✓ (2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correction explanation of (A)



## 69. Recognise diagram and choose correct option :



(1) A – Secondary  
B – Primary

(2) A – Secondary  
B – Tertiary

(3) A – Primary  
B – Secondary

☒ (4) A – Secondary  
B – Secondary

- Alpha helix
- Beta plated sheet
- Beta plated sheet
- Alpha helix
- Alpha helix
- Beta plated sheet
- Alpha helix
- Beta plated sheet





70. Given below are two statements :

**Statement I :** Each enzyme shows its highest activity at optimum temperature and optimum pH.

**Statement-II :** Enzyme isolated from "organism" who normal live under extremely high temperatures are stable and retain their catalytic power even at high temperature (upto 80°C-90°C).  
e.g

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ☒ (4) Both Statement I and Statement II are correct





**71. Amino acids are organic compounds containing an amino group and an acidic group on same \_\_\_\_\_ carbon**

(1) Beta

✓(2) Alpha

(3) Gamma

(4) Delta



72. Select the option which is not correct with respect to Co-factor.

- (1) Catalytic activity is lost when the co-factor is removed from the enzyme
- (2) Cofactor play a crucial role in the catalytic activity of the enzyme.
- ✓ (3) Three kinds of cofactors may be identified prosthetic groups, <sup>co-enzyme</sup> ~~apoenzymes~~ and metal ions.
- (4) Three kinds of cofactors may be identified: prosthetic groups, Co-enzymes and metal ions.



73. Which of the following represents uridylic acid :



(1) Uracil + Ribose

✓ (2) Uridine + Phosphoric acid

(3) Uracil + Phosphoric acid

(4) Uridine + Ribose + Phosphoric acid





74. Number of peptide bonds in "<sup>①</sup>dipeptide" and type of bond between nitrogen bases and sugar of nucleic acid are

(1) Two and N-glycosidic bonds

(2) One and ester bond

✓(3) One and N-glycosidic bonds

(4) Two and peptide bond



75. Nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin niacin are example of

(1) Prosthetic groups

✓ (2) Co-enzymes

(3) Metal ions.

(4) All of these



76. Which one of the following is incorrect match :

- |                          |                          |
|--------------------------|--------------------------|
| (1) Primary metabolite   | – Amino acids            |
| (2) Secondary metabolite | – Cellulose              |
| ✓ (3) GLUT-4             | – Fight infectious agent |
| (4) Alkaloids            | – Codeine                |





77. The number of fatty acid molecule which is present in mono, Di and Triglycerides respectively :

✓ (1) 1, 2, 3

(2) 0, 1, 2

(3) 1, 1, 1

(4) 2, 3, 4



78. Which of the following is not a protein :

(1) GLUT-4

(2) Collagen

✓ (3) Inulin = poly

(4) Antibody



79. Which of the following affect the activity of simple enzyme :

- (1) Temperature
- (2) pH
- (3) Substrate concentration
- ☒ (4) All of these





80. Few secondary metabolites and their examples are mentioned below. Select the incorrect pair

- |                        |   |                                                               |
|------------------------|---|---------------------------------------------------------------|
| (1) Alkaloids          | - | Morphine, Codeine                                             |
| (2) Drugs              | - | Vinblastin, Curcumin                                          |
| ✓(3) <del>Toxins</del> | - | <u>Concanavalin A</u> <sup>→ <u>lectin</u></sup> <sub>x</sub> |
| (4) Essential oils     | - | Lemon grass oil                                               |



81. Given below are two statements :

**Statement I :** Morphine, abrin, cellulose, Ricin are Secondary metabolite. ✓

**Statement-II :** We do not at the moment understand the role or function of all 'Secondary metabolite' in Host organism. ✓

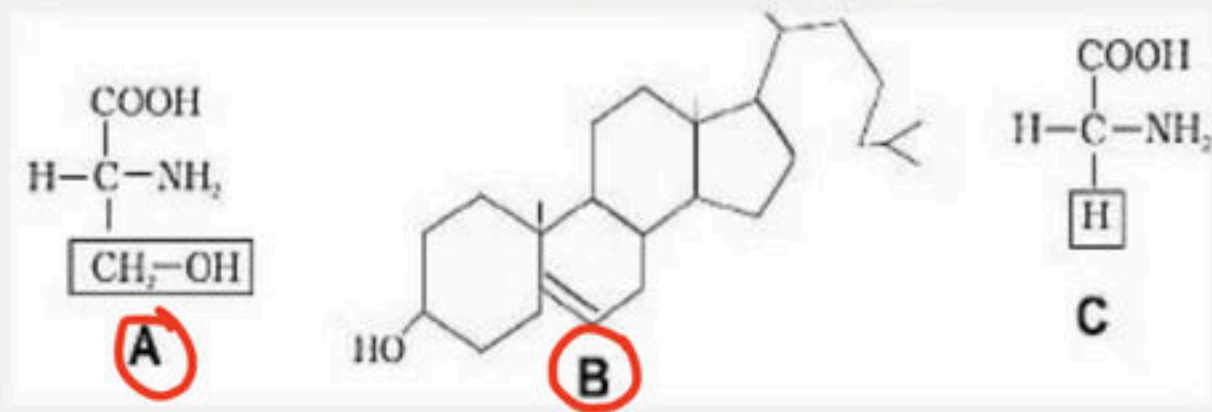
**Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct ✓





82. The given structures A,B,C are respectively



- (1) Alanine, cholesterol, serine
- (2) Glycine, lecithin, serine
- ☒ (3) Serine, cholestrol, glycine
- (4) Alanine, triglyceride, serine





**83. Given below are two statements :**

**Statement I :** Stability is something related to energy status of the molecule or the structure.

**Statement II :** During the state where substrate is bound to the enzyme active site, a new structure of substrate called transition state structure formed.

**Choose the correct answer from the options given below.**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ☒ (4) Both, Statement I and Statement II are correct





84. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A) :** Cellulose can not hold  $I_2$ .

**Reason (R) :** Cellulose does not contain complex helices.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- ✓ (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





85. Given below are two statements :

**Statement I :** Inorganic compound like sulphate, phosphate etc never seen in acid soluble fraction ✗

**Statement II :** Carbon and hydrogen is higher in any living organism than in earth's crust ✓

**Choose the correct answer from the options given below.**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- ✓ (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





86. Enzymes that catalyze the removal of groups from the substrate by mechanisms other than hydrolysis, leaving double bonds, belong to:

(1) Dehydrogenases

(2) Hydrolases

✓ (3) Lyases

(4) Transferases



87. The class of enzymes present in lysosomes is:

(1) Lyases

(2) Ligases

✓ (3) Hydrolases ✓

(4) Transferases



88. Which one is not a disaccharide?

(1) Lactose

(2) Sucrose

(3) Maltose

(4) Starch





**89. Glycosidic bond is formed between:**

- (1) Carbon and oxygen atoms of two adjacent monosaccharides
- (2) Carbon and hydrogen atoms of two adjacent monosaccharides
- (3) Hydrogen and oxygen atoms of two adjacent monosaccharides
- (4) Two carbon atoms of two adjacent monosaccharides



90. The bond between the phosphate and hydroxyl group of sugar is called:

- (1) Glycosidic bond
- (2) Phosphoanhydride bond
- ✓✓ (3) Ester bond
- (4) Peptide bond



**91. Which of the following groups contain all polysaccharides?**

- (1) Glycogen, sucrose, and maltose
- (2) Sucrose, glucose, and fructose
- (3) Maltose, lactose, and fructose
- ☒ (4) Glycogen, cellulose, and starch





92. Starch is a polymer of:

- (1) Glucose ↩
- (2) Fructose
- (3) Maltose
- (4) Sucrose



**93. Chitin is a polymer of:**

- (1) N-acetyl muramic acid
- (2) N-acetyl gluconic acid
- ☒ (3) N-acetyl glucosamine
- (4) None of the above



94. Find out the wrongly matched pair:

- (1) Primary metabolite — Ribose
- (2) Secondary metabolite — Anthocyanin
- (3) Protein — Insulin
- (4) Cellulose — ~~Heteropolymer~~

Homopolymer





95. Which of the following bond formation is dehydration synthesis?

- (1) Peptide bond ✓
- (2) Glycosidic bond ✓
- (3) Phosphodiester bond ✓
- (4) All of the above ✓



**96. Inulin is a polymer of:**

- (1) Glucose
- ✓ (2) Fructose
- (3) Galactose
- (4) Sucrose



97. Which of the following groups represents only primary metabolites?

- (1) Glucose, ribose, glycine, alanine, cholesterol
- (2) Lecithin, serine, glycerol, palmitic acid, nucleotide
- (3) Adenosine, nucleosides, nitrogen bases, adenylic acid, triglyceride
- (4) All of the above

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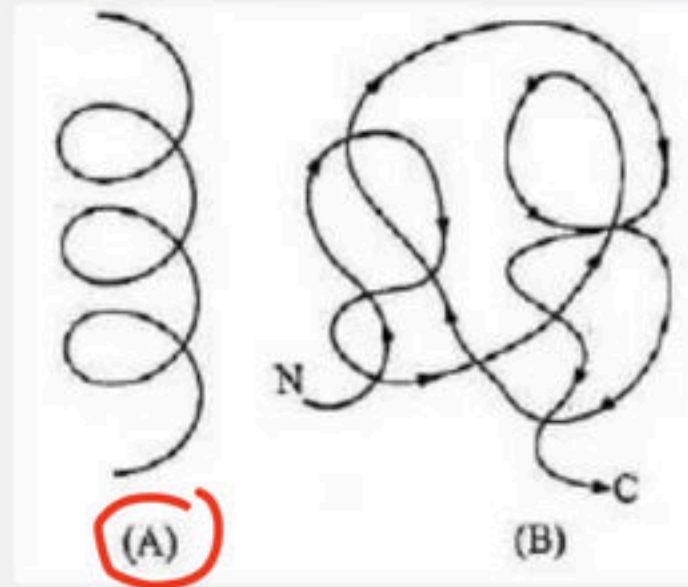
98. A protein is imagined as a line; the left end is represented by the first amino acid, and the right end is represented by the last amino acid.

The first and last amino acids are called as:

- ✓ (1) N-terminal amino acid and C-terminal amino acid, respectively
- (2) C-terminal amino acid and N-terminal amino acid, respectively
- (3) O-terminal amino acid and C-terminal amino acid, respectively
- (4)  $\text{NH}_2$ -terminal amino acid and  $\text{COOH}$ -terminal amino acid, respectively



99. Recognize the figure and find out the correct matching:



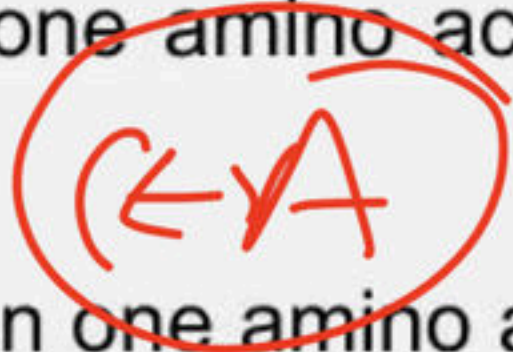
- (1) (A)—Primary structure; (B)—Secondary structure
- (2) (A)—Secondary structure; (B)—Primary structure
- ☒ (3) (A)—Secondary structure; (B)—Tertiary structure
- (4) (A)—Tertiary structure; (B)—Quaternary structure





**100. Peptide bond is formed when:**

- (1) Carboxyl group of one amino acid reacts with the carboxyl group of the next amino acid
- (2) Amino group of one amino acid reacts with the amino group of the next amino acid
- ✓ (3) Carboxyl group on one amino acid reacts with the amino group of the next amino acid
- (4) Amino group on one amino acid reacts with the carboxyl group of the next amino acid



X





THANK YOU



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Thank  
You

